8 A particle moves with a constant angular velocity ω in a circle of radius r centered at point O.

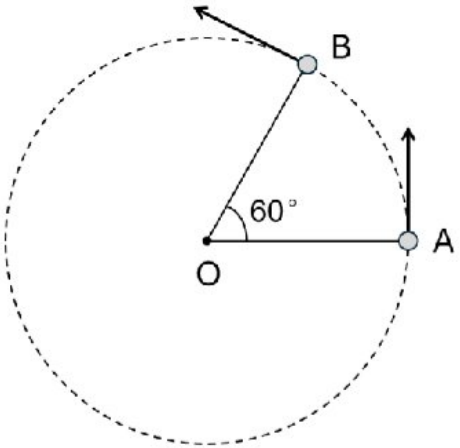

Diagram not drawn to scale

What is the magnitude of the change in velocity between points A and B?

A zero **B** $\frac{1}{2}r\omega$ **C** $r\omega$ **D** $2\,r\omega$